Hamming Code

(*) A fundamental type of Channel Coding.

(*) Linear Block coding

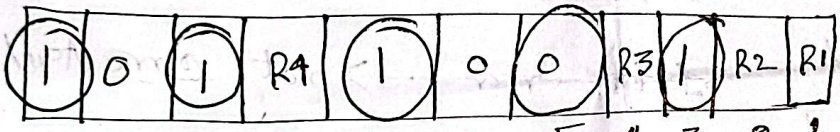
(*) Message is divided into fixed-sized blocks.

(*) $2^r \geq m+r+1$

Even parity

Odd parity

1011001 , $m=7$, $r=4$, $2^r = 16 \geq 7+4+1 = 12$



11 10 9 8 7 6 5 4 3 2 1

2^3 2^2 2^1 2^0

0 1 1 0

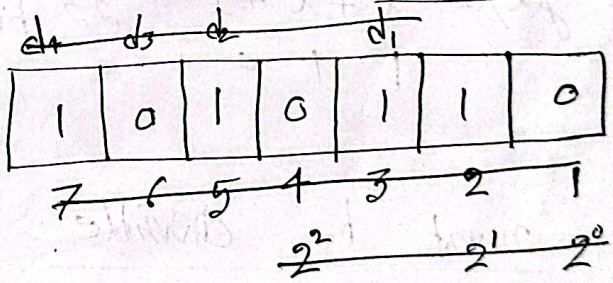
0 0 1 1
0 0 0 0
0 0 1 1
0 1 0 0
1 0 1 0
0 1 1 0

0 1 1 1
1 0 0 0
1 0 0 1
1 0 1 0
1 0 1 1

R1 = 1, 3, 5, 7, 9, 11

even parity, R1 = 0

Error Detection and Correction



1 2 3 4 5 6 7

2^0 2^1 2^4

0 1 1 0 0 1 1

P1 P2 D1 P4 D2 D3 D4 P0

1 2 3 4 5

0 1 [redacted] 1 1 1

1 2 3 4 5 6 7

2^0 2^1 2^4

2^{i+1}

24B → 80 CRC

24B → 120 HC

$S_1 = 1$

$S_2 = 0$

$S_4 = 1$

$(101)_2 = 5$, flip bit at pos 5

SECDED

Po at the end

S	P0	
000	correct	no-error
≠ 000	incorrect	1-bit error
≠ 000	correct	>1-bit error → not correctable
000	incorrect	error in parity

100100

CRC-32

32-degree polynomial

$g(x) = x^3 + x^2 + 1$ 33 bit string

1 1 0 1

100100 — 3 zeros / 3 order

must be divisible

24 —— 51 → CRC

121 → HC

Low-Density Parity-Check

LDPC

(i) Type of error-correction code.

(ii) It is a linear block coding technique.

(iii) Code word = data word + parity bits
redundant bits

(iv) Memoryless - encoding current block does not depend on any previous block.

(v) Matrix-based system. → (4-1) × 7

The Definitions (7,4)

⇒ n (Block Length) → code word length

⇒ k → data word length

⇒ m → (n-k) → the number of parity bits

Code rate, $R = \frac{k}{n}$

High rate (e.g., 8/9) → the connection is very good.

Low rate (e.g., 1/2, 1/3) → the environment is very noisy.

Parity-Check Matrix:

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$$

Validity
Reliability
Flexibility
Fairness

Student Assessment

~~The~~ How to generate H.

(i) Random Construction (Gallagers Method)

(ii) Quasi-Cyclic Construction.

(iii) Progressive Edge-Growth (PEG).

(iv) Finite Geometry Construction

For (7,4) → 3×7 ⇒ [P/I]

For our simple take an identity matrix at the right 3×3 part and take the other values randomly.

Let, $H = \left[\begin{array}{cccc|ccc} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{array}\right]$

Data word → 1011

1. $P1 \oplus \overset{1}{d_1} \oplus \overset{0}{d_2} \oplus \overset{1}{d_4} = 0 \Rightarrow P1 \oplus 1 \oplus 0 \oplus 1 = 0$

$\Rightarrow P1 = 0$

2. $P2 \oplus \overset{1}{d_1} \oplus \overset{1}{d_3} \oplus \overset{1}{d_4} = 0 \Rightarrow P2 \oplus 1 \oplus 1 \oplus 1 = 0$

$\Rightarrow P2 = 1$

3. $P3 \oplus \overset{0}{d_2} \oplus \overset{1}{d_3} \oplus \overset{1}{d_4} = 0 \Rightarrow P3 \oplus 0 \oplus 1 \oplus 1 = 0$

$\Rightarrow P3 = 0$

Code word = 1011010

24B 80 CRC →

24B 120 HC →

24B 23-80 En-LDPC → 7

24B

Decode

1011	010
data word	parity part.

Simple

Error checking

$$He^T = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}$$

Syndrome

$$= \begin{bmatrix} 1 \oplus 0 \oplus 0 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \\ 1 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 1 \oplus 0 \\ 0 \oplus 0 \oplus 1 \oplus 1 \oplus 0 \oplus 0 \oplus 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Ex

DW	P
1011	010
0111	001

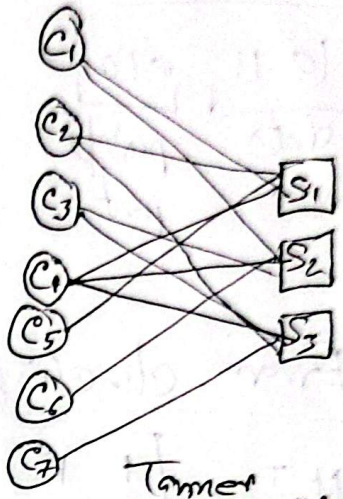
Error Correction

1011010

0011010 → wrongly received

Syndromes

$$\begin{bmatrix} c_1 & c_2 & c_3 & c_4 & c_5 & c_6 & c_7 \\ 1 & 1 & 0 & 1 & 0 & 0 & \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$



Tanner Graph (Bipartite)

Here c_1 appears in 2 wrong eqns, so check again after flipping.

Pros

(i) Extreme Accuracy

(ii) High Speed (parallelism)

(iii) Efficiency

(iv) ~~Variable~~

Cons

(i) Encoding Complexity

(ii) Latency - short message, not good.